



IIT ROORKEE



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CERTIFICATION COURSE

Monopolistic Competition and Trade

Dr. PRATAP C. MOHANTY

DEPT. OF HUMANITIES & SOCIAL SCIENCES, IIT ROORKEE



The “Krugman Model”: Assumptions

1. **Internal economies of scale**
2. **Monopolistic competition** (non-homogeneous goods)
3. One factor of production (labour)
4. Identical preferences
5. Large number of goods produced with the same technology
6. Full employment

Paul R. Krugman (1979): Increasing returns, monopolistic competition, and international trade. *Journal of International Economics*, Vol. 9(4): 469-479

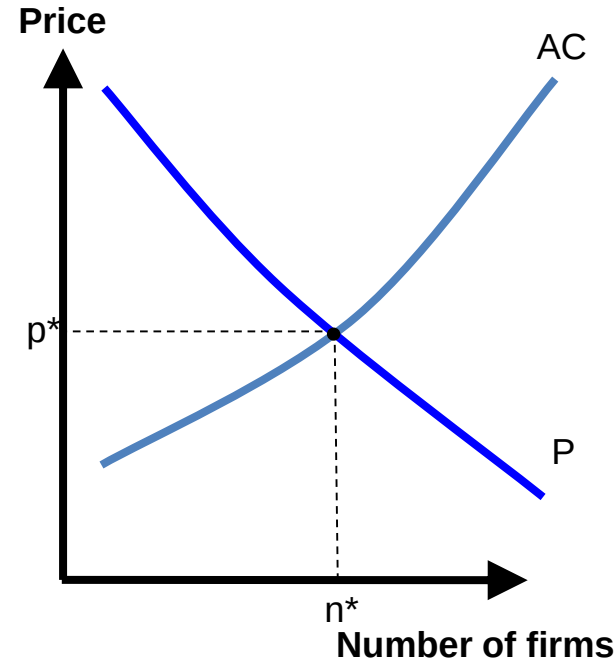
Key assumption: Economies of Scale

- External: cost per unit depends on the size of the industry, not the firm (Silicon Valley, Hollywood...)
 - **Internal: cost per unit depends on the size of the firm**, not industry (Nokia, Phillips, GE...)
 - Krugman models technology as
$$L = a + b*Q \Leftrightarrow Q = 1/b*L - a/b$$
 - the amount of labour required (L) to produce amount of input (Q) depends on b *and* constant a (fixed cost)
- ➔ Doubling the inputs *more than* doubles the output (increasing internal economies of scale)

Long-Run Market Equilibrium of Monopolistically Competitive Market

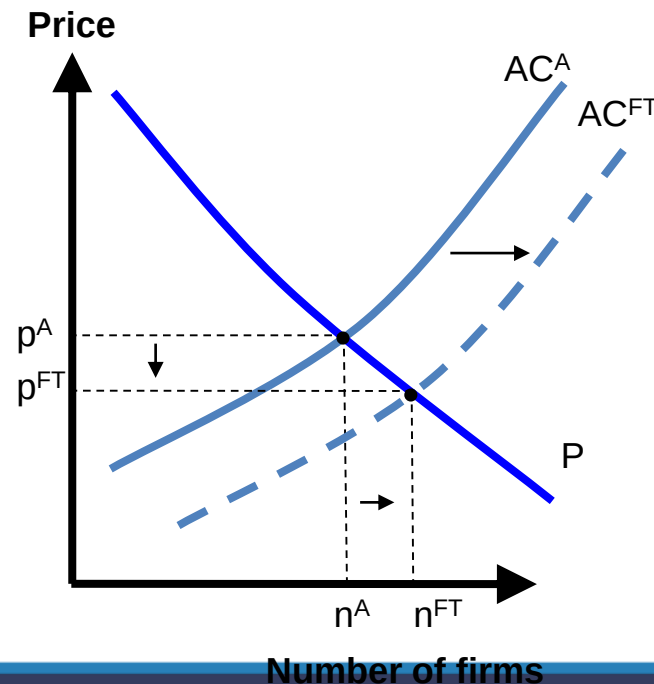
The more firms there are:

1. the less each firm produces $\rightarrow$ higher average cost (due to increasing returns to scale) $\rightarrow$ upward sloping cost curve
2. the harder the competition $\rightarrow$ decreasing price $\rightarrow$ downward sloping price curve



Introducing Trade to the Monopolistic Competition Model

- **Trade increases market size**
→ firms exploit more of the returns to scale → average cost decreases → price decreases → number of firms increases
- i.e. a larger variety of products is available for smaller price
- *everybody* are better off even if the countries are identical



Monopolistic Competition

- **Monopolistic competition** is a model of an imperfectly competitive industry which assumes that
 1. Each firm can differentiate its product from the product of competitors.
 2. Each firm ignores the impact that changes in its own price will have on the prices of competitors set: even though each firm faces competition it behaves as if it were a monopolist.

- A firm in a monopolistically competitive industry is expected:
 - to sell more the larger the total sales of the industry and the higher the prices charged by its rivals.
 - to sell less the larger the number of firms in the industry and the higher its own price.
- These concepts are represented by the mathematical relationship:

$$Q = S[1/n - b(P - P^*)]$$

- Q is an individual firm's sales
- S is the total sales of the industry
- n is the number of firms in the industry
- b is a constant term representing the responsiveness of a firm's sales to its price
- P is the price charged by the firm itself
- P^* is the average price charged by its competitors

- To make the model easier to understand, we assume that all firms have identical demand functions and cost functions.
 - Thus in equilibrium, all firms charge the same price: $P = P$
- In equilibrium,
 - $Q = S/n + 0$
 - $AC = C/Q = F/Q + c = F(n/S) + c$

1. *The Number of Firms and Average Cost*

$$AC = F(n/S) + c$$

- The larger the number of firms n in the industry, the higher the average cost for each firm because the less each firm produces.
- The larger the total sales S of the industry, the lower the average cost for each firm because the more that each firm produces.

2. *The Number of Firms and the Price*

$$Q = S[1/n - b(P - P^*)]$$

$$Q = S/n - Sb(P - P^*)$$

$$Q = S/n + SbP - SbP^*$$

$$Q = A - BP$$

- Let $A \equiv S/n + SbP$ and $B \equiv Sb$

$$MR = P - Q/B = c$$

$$MR = P - Q/Sb = c$$

$$P = c + Q/Sb$$

$$P = c + (S/n)/Sb$$

$$P = c + 1/(nxb)$$

- The larger the number of firms n in the industry, the lower the price each firm charges because of increased competition.

- At some number of firms, the price that firms charge (which decreases in n) matches the average cost that firms pay (which increases in n).
- This number of firms is the number at which each firm has *zero profits*: price matches average cost.
- This number is the equilibrium number of firms.

- If the number of firms is greater than or less than n_2 , then the industry is not in equilibrium in the sense that firms have an incentive to exit or enter the industry.
 - Firms have an incentive to enter the industry when profits are greater than zero (price > average cost).
 - Firms have an incentive to exit the industry when profits are less than zero (price < average cost).

- Because trade increases market size, trade is predicted to decrease average cost in an industry described by monopolistic competition.
 - Industry sales increase with trade leading to decreased average costs:
 $AC = F(n/S) + c$
- Because trade increases the variety of goods that consumers can buy under monopolistic competition, it increases the welfare of consumers.
 - Because average costs decrease, consumers can also benefit from a decreased price.

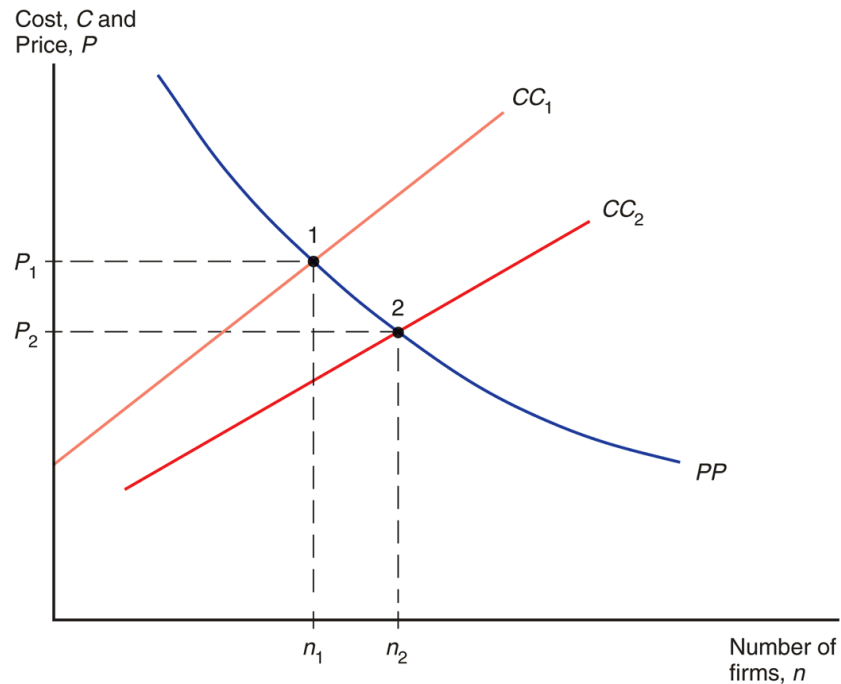


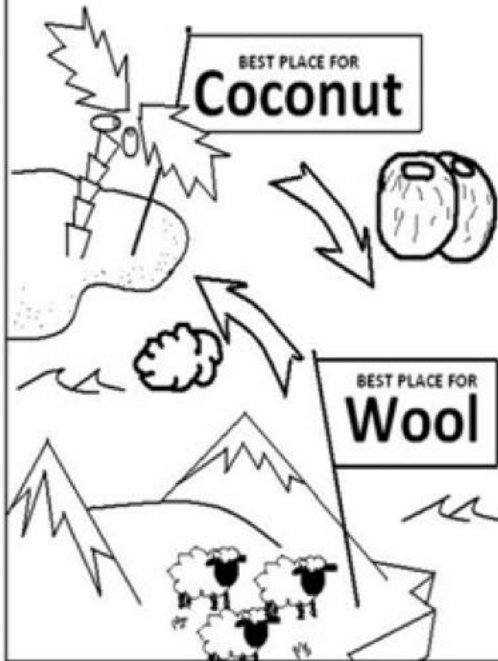
Figure 6-4

Effects of a Larger Market

An increase in the size of the market allows each firm, other things equal, to produce more and thus have lower average cost. This is represented by a downward shift from CC_1 to CC_2 . The result is a simultaneous increase in the number of firms (and hence in the variety of goods available) and fall in the price of each.

OLD TRADE THEORY

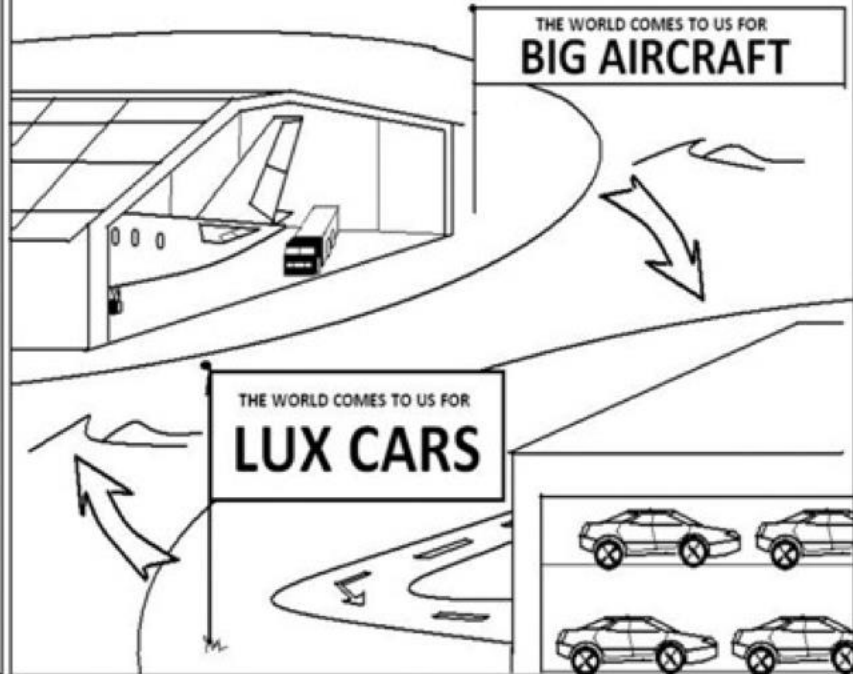
(BASED ON COMPARITIVE ADVANTAGE)



we're good at different things,
and that's why we trade...

NEW TRADE THEORY

(BASED ON 'ECONOMIES OF SCALE',
OR ' THINGS CAN BE CHEAPER IF THE
SCALE OF OPERATION IS LARGE')



we're similar in abilities ; each of us picks
a good requiring large scale production

Intra- and Inter-industry Trade

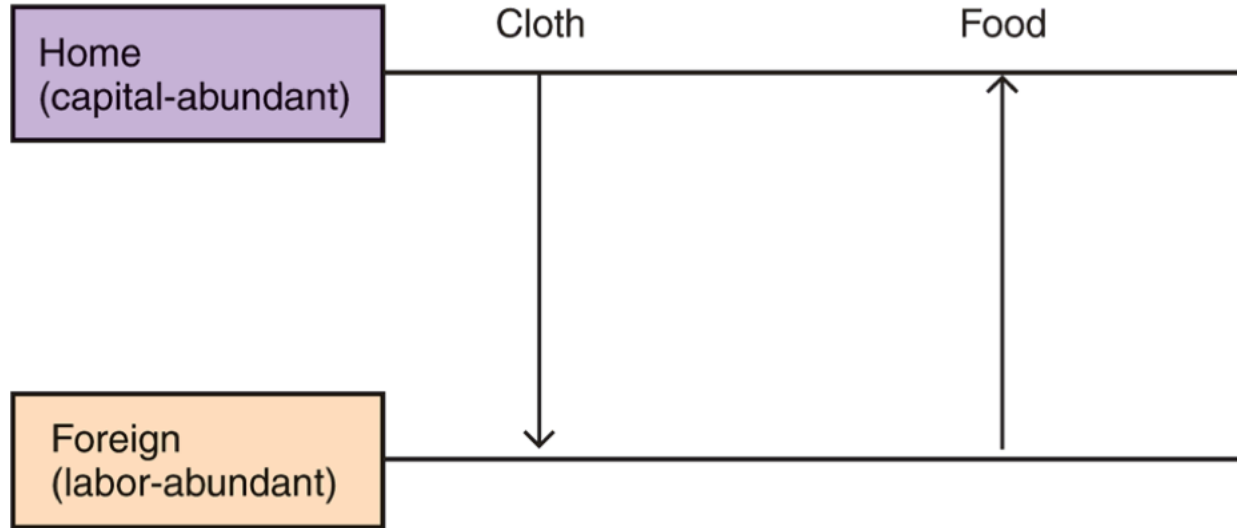
- **Inter-industry trade:** countries export goods of one product category and imports goods of other product category as in the **Ricardian** and the **Heckscher-Ohlin** models
 - Basis for trade: **Comparative Advantage** due to differences in productivity (Ricardian model) or in factor endowments (HO-model)
- **Intra-industry trade:** countries export and import products of the same products category as in the **Krugman model**
 - Basis for trade: **Internal economies of scale**

Inter-industry Trade

Figure 6-6

Trade in a World Without Increasing Returns

In a world without economies of scale, there would be a simple exchange of cloth for food.



Intra-industry Trade

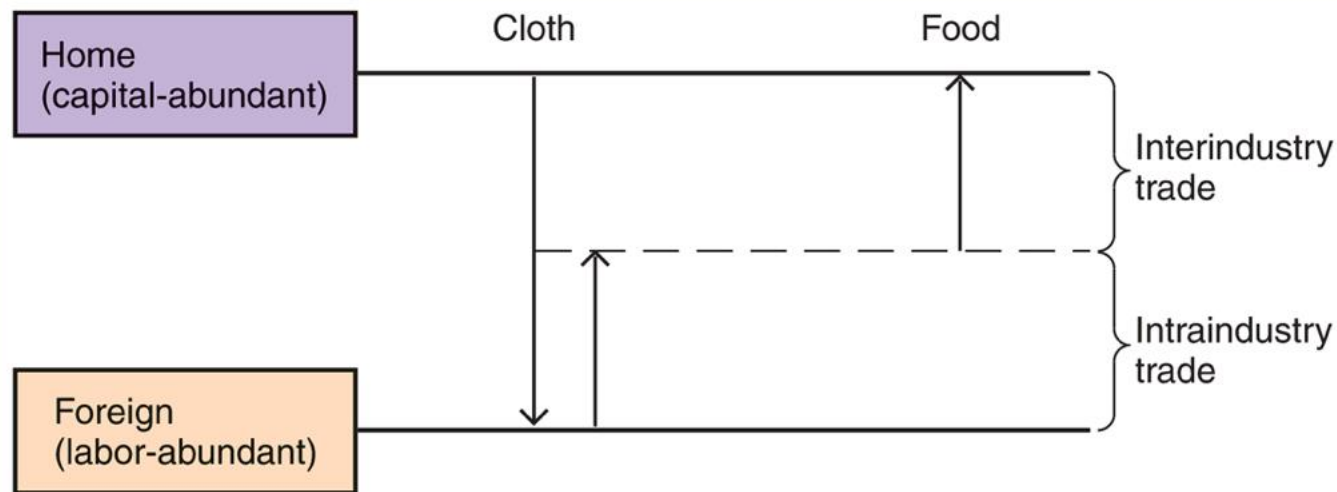


Figure 6-7

Trade with Increasing Returns and Monopolistic Competition

If cloth is a monopolistically competitive industry, Home and Foreign will produce differentiated products. As a result, even if Home is a net exporter of cloth goods, it will import as well as export cloth, giving rise to intraindustry trade.

Characteristics of IIT

- Intra-industry trade between industrial countries is common
- Fundamental problem is defining an industry.
- For example, if computers are defined as office machinery, then computers and pencil sharpeners are in the same industry,
- More broadly an industry is defined, the more trade appears to be intra-industry
- Evidence suggests that intra-industry trade is greater –in high technology industries –where there is more scope for product differentiation –in countries more open to trade –in nations that have received larger amounts of foreign direct investment

Type of Trade	Phrase	Meaning	Source
<i>Inter-industry</i>	Either/or	Either imports or exports in a given sector of the economy	Comparative advantage
<i>Intra-industry</i>	Both/and	Both imports and exports in a given sector of the economy	Product differentiation

Gains from IIT

- **Lower prices** : An increase in the size of the market allows for scale economies, which lowers production costs and eventually prices to consumers
- **Increase in the number of firms** : There is a high likelihood that intra industry trade expands the number of domestic firms and the quantity of domestic output
- **Increase in consumer choices** : Intra- industry trade tends to give access to a much greater variety of goods than produced domestically

Models with Heterogeneous Firms

Melitz (2003, Econometrica) model

- **Extension of the “Krugman model”**
- firms can enter an industry by paying a fixed **entry cost**, they then learn their productivity (profitability) and leave if this is too low
- There are fixed and variable costs also for exporting → only the most productive firms export
- reductions in barriers to trade → increase profits of exporters and reduce the export productivity cutoff → labor demand within the industry rises → increase in wages → profits of nonexporter decrease → less productive firms bankrupt

A new source of gains from trade: Increase in **productivity**

*Thank
you*

